DAY 3 - Assignment.

1)

a) $F(A,B,C) = \bar{A}C + B\bar{C}$

convert into canonical:

$$\bar{A}C(B+\bar{B}) + B\bar{C}(A+\bar{A})$$

$$\bar{A}BC + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$$

TT:

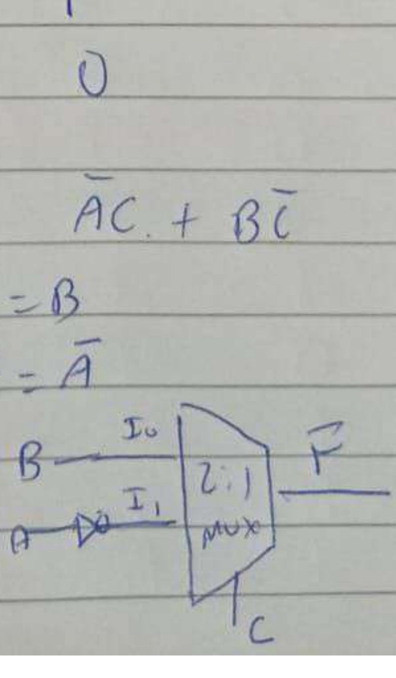
~~Sum~~

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

i) looking at eq^n : $\bar{A}C + B\bar{C}$

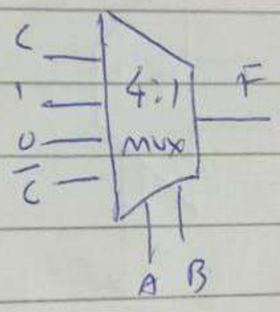
If $C=0$: $F = B$

$C=1$: $F = \bar{A}$



ii)

A	B	F
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$



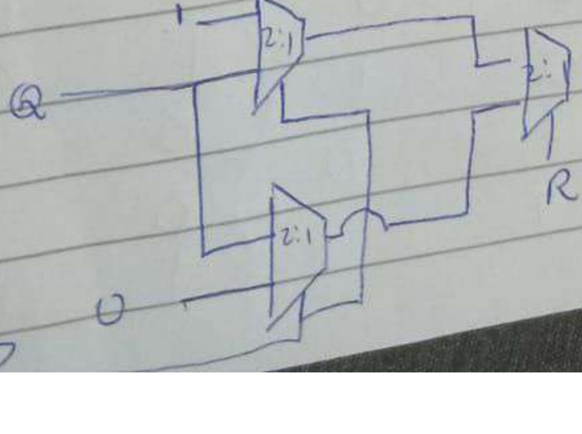
b) $\bar{P}\bar{R} + Q\bar{R} + \bar{P}Q = F(P, Q, R)$

$$F(P,Q,R) = \bar{P}\bar{R}Q + \bar{P}\bar{R}\bar{Q} + \cancel{\bar{P}\bar{R}Q} + P\bar{R}Q + \bar{P}RQ + \cancel{\bar{P}\bar{R}Q}$$

$$\Rightarrow \bar{R}(\bar{P}Q + \bar{P}\bar{Q} + PQ) + R(\bar{P}Q)$$

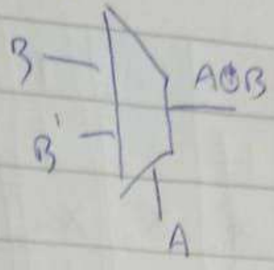
$$\bar{R}(\bar{P} + Q) + R(\bar{P}Q)$$

taking R as select lines:

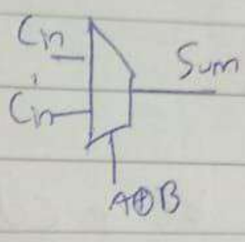


For Sum: using 2MUX

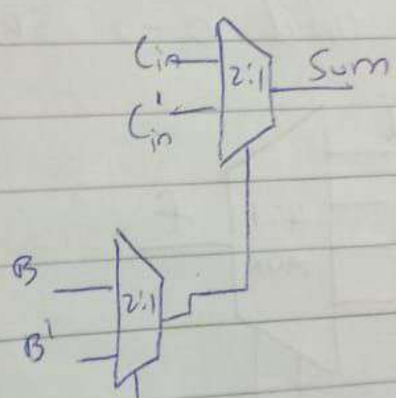
MUX1: for $A \oplus B$



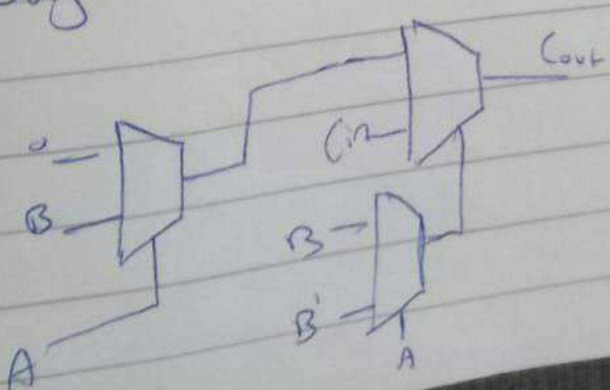
MUX2: $A \oplus B \oplus C_{in}$



Finally:

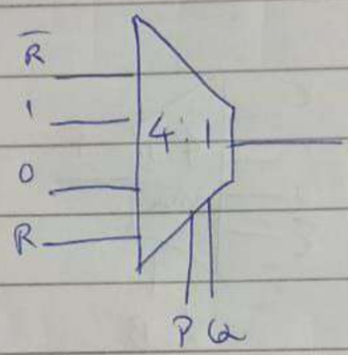


For carry:



ii)

P	Q	F
0	0	$\bar{R}$
0	1	1
1	0	0
1	1	R



2) FA using 2:1 MUX

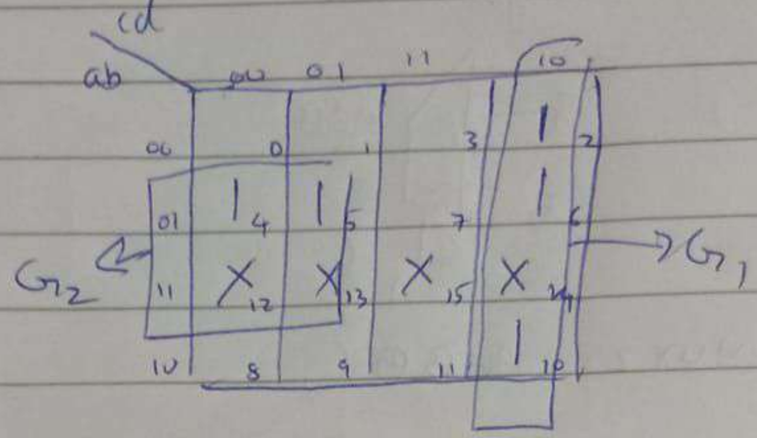
S, C (A, B, C_{in})

$S = A \oplus B \oplus C_{in}$

A	B	C_{in}	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

3) $f(a,b,c,d) = \Sigma(2,4,5,6,10) + D(12,13,14,15)$

Sol^n



using k-map:

$$f = b\bar{c} + c\bar{d}$$

$\Rightarrow \cancel{ab\bar{c} + \bar{a}b\bar{c}}$

let a and b → select lines